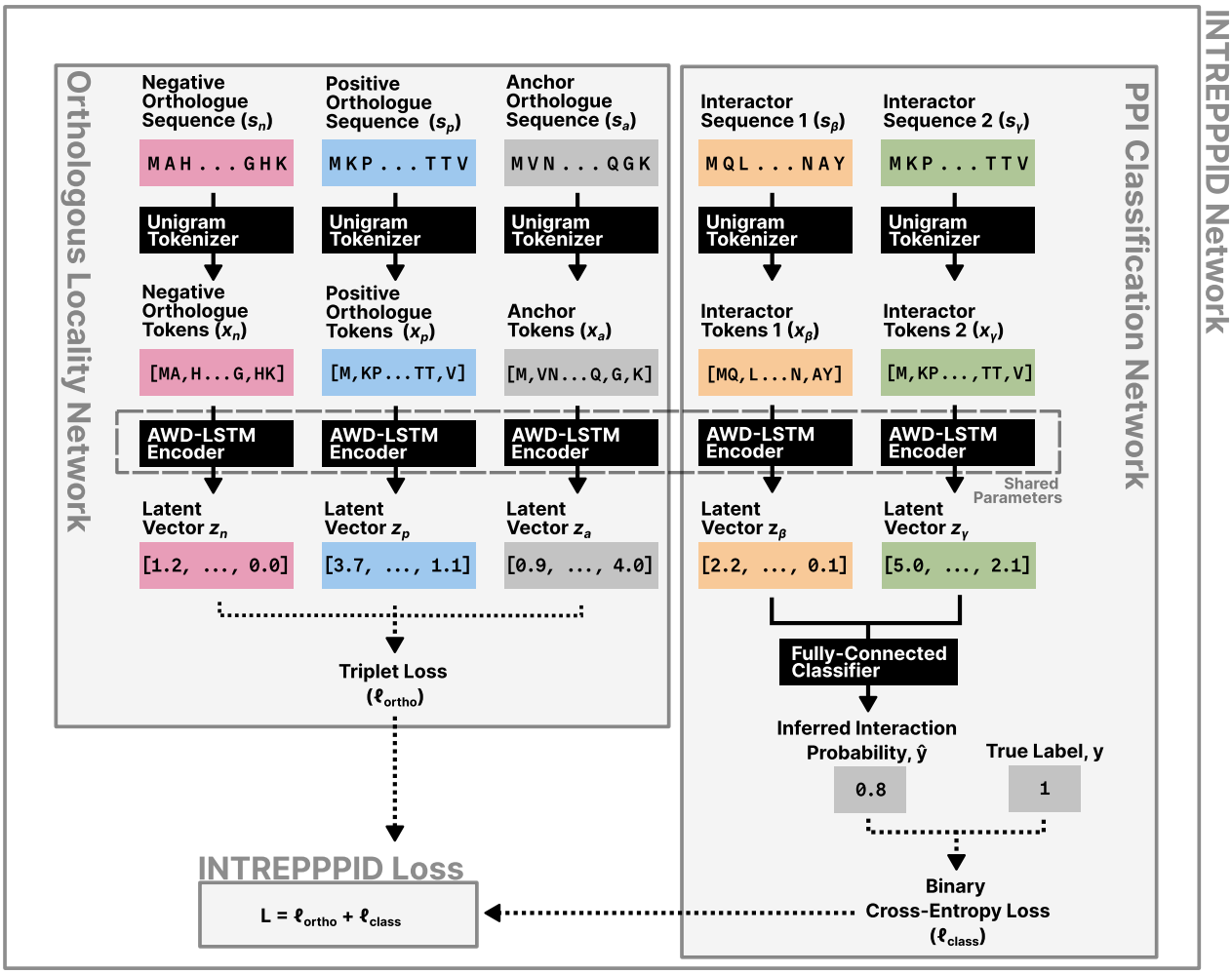
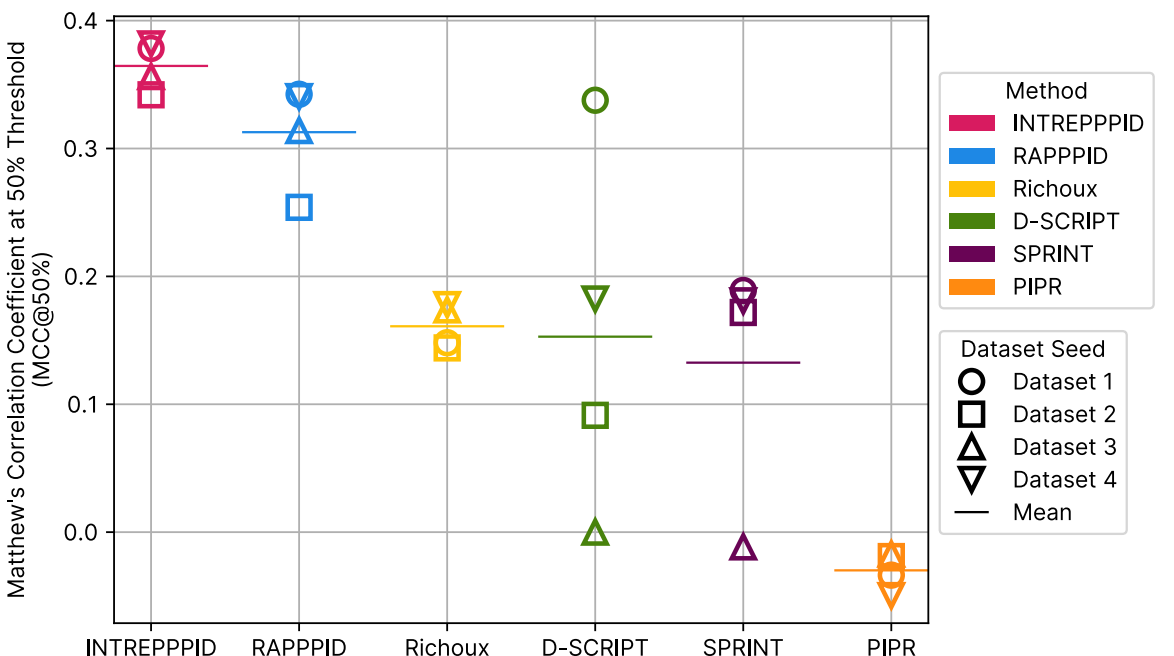
A)


INTREPPPID Network
Orthologous Locality Network
Negative Orthologue Sequence (s_n)
Positive Orthologue Sequence (s_p)
Anchor Orthologue Sequence (s_a)
MAH...GHK
MKP...TTV
MVN...QGK
Unigram Tokenizer
Negative Orthologue Tokens (x_n)
Positive Orthologue Tokens (x_p)
Anchor Tokens (x_a)
[MA,H...G,HK]
[M,KP...TT,V]
[M,VN...Q,G,K]
AWD-LSTM Encoder
Shared Parameters
Latent Vector z_n
Latent Vector z_p
Latent Vector z_a
[1.2, ..., 0.0]
[3.7, ..., 1.1]
[0.9, ..., 4.0]
Triplet Loss (ℓ_{ortho})
PPI Classification Network
Interactor Sequence 1 (s_β)
Interactor Sequence 2 (s_γ)
MQL...NAY
MKP...TTV
Interactor Tokens 1 (x_β)
Interactor Tokens 2 (x_γ)
[MQ,L...N,AY]
[M,KP...,TT,V]
Latent Vector z_β
Latent Vector z_γ
[2.2, ..., 0.1]
[5.0, ..., 2.1]
Fully-Connected Classifier
Inferred Interaction Probability, ŷ
True Label, y
0.8
1
Binary Cross-Entropy Loss (ℓ_{class})
INTREPPPID Loss
L = ℓ_{ortho} + ℓ_{class}

B)


Matthew's Correlation Coefficient at 50% Threshold (MCC@50%)
0.4
0.3
0.2
0.1
0.0
INTREPPPID
RAPPPID
Richoux
D-SCRIPT
SPRINT
PIPR
Method
INTREPPPID
RAPPPID
Richoux
D-SCRIPT
SPRINT
PIPR
Dataset Seed
Dataset 1
Dataset 2
Dataset 3
Dataset 4
Mean